



Modelling Financial Markets

Seminar 8: Backtesting and Evaluating Risk Forecasts — Hands-on



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Acronyms used in this material

- ▣ ACI: Adaptive Conformal Inference
- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ DM: Diebold–Mariano (test)
- ▣ ES: Expected Shortfall
- ▣ FHS: Filtered Historical Simulation
- ▣ FRTB: Fundamental Review of the Trading Book
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-EVT: GARCH filter combined with Extreme Value Theory for the tail
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HS: Historical Simulation
- ▣ MCS: Model Confidence Set

Outline

Introduction

Key Formulas

Part A: Pen and Paper

Part B: Computation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- Compute the Kupiec, Christoffersen and Basel traffic-light statistics step by step
- Backtest ES with residuals and the Acerbi–Szekely Z_2 by simulation
- Rank (VaR, ES) forecasts with the FZ0 loss, Diebold–Mariano tests and the MCS
- Build a conformal VaR and check its coverage in calm and crisis periods

- Background: Lecture 8; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- **Part A:** nine pen-and-paper exercises (A1–A9)
- **Part B:** eight computational exercises with inference (B1–B8)
- **Part C:** an open question, a seed for the team project (C1), and the critique of an AI answer (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Report every test with its p -value and say how it was obtained (asymptotic, exact, bootstrap, simulation)
- A model that passes a test is not validated: ask how much power the test had

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (Kupiec), A3 (Christoffersen), A5 (traffic light), A6 (FZ0), A7 (Z_1 , Z_2), A8 (conformal quantile, ACI)
- Part B: B1 (VaR tests, S&P 500), B3 (ES tests with simulation), B5 (Diebold–Mariano with HAC), B7 (conformal VaR in crises)

Proposed: you solve them

- Part A: A2 (model: A1), A4 (model: A3), A9 (model: A6)
- Part B: B2 (model: A5, B1), B4 (model: B3), B6 (model: B5), B8 (model: B7)
- Part C: C1, open question and project seed; C2, critique of an AI answer

Setup: Python Environment and Data

-  **Open in Colab:** `chapter8_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `arch`, `matplotlib`
- Series: S&P 500, BET, Bitcoin (daily closes); EUR/RON: BNR reference rate; up to 18 September 2026
- Forecasts: the six models of Lecture 8 (HS, Normal, Student-t, GARCH-t, FHS, GARCH-EVT), rolling window of 1000 days, parameters re-estimated every 20 days
- Losses: $L_t = -r_t$ with $r_t = \ln P_t - \ln P_{t-1}$; breach: $I_t = 1\{L_t > \text{VaR}_t\}$

Key Formulas (1/2): VaR Backtests

- Kupiec [Kupiec \(1995\)](#): $LR_{uc} = -2 [(T - x) \ln(1 - p) + x \ln p - (T - x) \ln(1 - \hat{\pi}) - x \ln \hat{\pi}]$, $\hat{\pi} = x/T$, χ_1^2
- Christoffersen [Christoffersen \(1998\)](#): $\hat{\pi}_{01} = \frac{n_{01}}{n_{00} + n_{01}}$, $\hat{\pi}_{11} = \frac{n_{11}}{n_{10} + n_{11}}$, $\hat{\pi} = \frac{n_{01} + n_{11}}{n}$
- $LR_{ind} = -2 [(n_{00} + n_{10}) \ln(1 - \hat{\pi}) + (n_{01} + n_{11}) \ln \hat{\pi} - n_{00} \ln(1 - \hat{\pi}_{01}) - n_{01} \ln \hat{\pi}_{01} - n_{10} \ln(1 - \hat{\pi}_{11}) - n_{11} \ln \hat{\pi}_{11}]$
- $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi_2^2$; durations [Christoffersen & Pelletier \(2004\)](#): Weibull shape $b = 1$ under H_0
- Traffic light [BCBS \(1996\)](#): $X \sim \text{Binomial}(250, 0.01)$; yellow from $P(X \leq x) \geq 95\%$, red from $P(X \leq x) \geq 99.99\%$

Key Formulas (2/2): ES, Scores and Conformal VaR

- McNeil–Frey [McNeil & Frey \(2000\)](#): $e_t = (L_t - \text{ES}_t)/\sigma_t$ on breach days; $H_0: E[e_t] = 0$, $H_1: E[e_t] > 0$
- Acerbi–Szekely [Acerbi & Szekely \(2014\)](#): $Z_1 = 1 - \frac{1}{N_T} \sum_t \frac{L_t l_t}{\text{ES}_t}$, $Z_2 = 1 - \frac{1}{T\alpha} \sum_t \frac{L_t l_t}{\text{ES}_t}$; fixed thresholds -0.70 and -1.8 for $T = 250$
- FZ0 [Patton, Ziegel & Chen \(2019\)](#): $\frac{\mathbf{1}\{L > v\}(L-v)}{\alpha e} + \frac{v}{e} + \ln e - 1$, with VaR v and ES e (positive losses)
- Diebold–Mariano [Diebold & Mariano \(1995\)](#): $\bar{d}/\sqrt{\widehat{\text{LRV}}/T}$ with Newey–West weights [Newey & West \(1987\)](#); MCS [Hansen, Lunde & Nason \(2011\)](#)
- Conformal VaR: $\hat{\sigma}_t s_{(k)}$, $k = \lceil (n+1)(1-\alpha) \rceil$; ACI [Gibbs & Candès \(2021\)](#): $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$

A1: Kupiec Test, Step by Step [Solved]

Task

- A VaR 1% model produced $x = 7$ breaches in $T = 250$ days
- (a) Expected number of breaches and $\hat{\pi}$
- (b) $\ln L_0$, $\ln L_1$ and LR_{uc}
- (c) Decision at 5%; compare with the exact probability $P(X \geq 7)$ under $\text{Binomial}(250, 0.01)$

Solution

- Expected $250 \cdot 0.01 = 2.50$; $\hat{\pi} = 7/250 = 0.028$
- $\ln L_0 = 243 \ln 0.990 + 7 \ln 0.010 = -34.678$
- $\ln L_1 = 243 \ln 0.972 + 7 \ln 0.028 = -31.930$
- $LR_{uc} = -2(-34.678 - (-31.930)) = 5.497 > 3.84$,
 $p = 0.019$: reject
- Exact one-sided $P(X \geq 7) = 0.0137$: same conclusion; in the traffic light, 7 exceptions is yellow

A2: Kupiec for VaR 2.5% [Proposed]

Solution discussed in class

Task

- A VaR 2.5% (the FRTB desk level) was breached $x = 11$ times in $T = 250$ days
- (a) Expected breaches, $\hat{\pi}$, LR_{UC} and p -value
- (b) Is the desk limit of 30 exceptions relevant here? What does that say about the limit?
- Model: A1

A3: Christoffersen Transition Counts [Solved]

Task

- ▣ BET, VaR 1% of HS, 2007–2010 (1005 days):
 $n_{00} = 958$, $n_{01} = 22$, $n_{10} = 22$, $n_{11} = 2$
- ▣ (a) $\hat{\pi}_{01}$, $\hat{\pi}_{11}$, $\hat{\pi}$
- ▣ (b) LR_{ind} and its p -value
- ▣ (c) LR_{cc} , given $LR_{uc} = 14.11$

Solution

- ▣ $\hat{\pi}_{01} = 22/(958 + 22) = 0.0224$,
 $\hat{\pi}_{11} = 2/(22 + 2) = 0.0833$, $\hat{\pi} = 0.0239$
- ▣ $LR_{ind} = 2.32$, $p = 0.128$: independence not rejected,
although $\hat{\pi}_{11} \approx 4 \hat{\pi}_{01}$
- ▣ $LR_{cc} = 14.11 + 2.32 = 16.43$, $p = 0.0003$: rejected,
driven by coverage
- ▣ Only 2 pairs of consecutive breaches: too few to estimate
 π_{11} precisely

A4: Christoffersen for GARCH-t [Proposed]

Solution discussed in class

Task

- ▣ Same period and index, VaR 1% of GARCH-t: $n_{00} = 966$, $n_{01} = 19$, $n_{10} = 19$, $n_{11} = 0$
- ▣ (a) $\hat{\pi}_{01}$, $\hat{\pi}_{11}$, $\hat{\pi}$ and LR_{ind} (use $0 \ln 0 = 0$)
- ▣ (b) LR_{uc} , LR_{cc} and the decision at 5%
- ▣ (c) Which model has the better conditional coverage, HS (A3) or GARCH-t?
- ▣ Model: A3

A5: The Traffic Light from the Binomial Distribution [Solved]

Task

- X = exceptions of a correct VaR 1% in 250 days, $X \sim \text{Binomial}(250, 0.01)$
- (a) Compute $P(X \leq x)$ for $x = 4, 5, 9, 10$ and derive the zones
- (b) A bank reports 6 exceptions: zone, 1996 plus factor, FRTB multiplier
- (c) Probability that a model with a true 97% coverage still lands in the green zone

Solution

- $P(X \leq 4) = 89.22\%$, $P(X \leq 5) = 95.88\%$,
 $P(X \leq 9) = 99.97\%$, $P(X \leq 10) = 99.99\%$
- Green 0–4 (below 95%), yellow 5–9, red from 10 (at least 99.99%)
- 6 exceptions: yellow; plus factor 0.50 (1996); multiplier 1.76 (FRTB); $P(X \geq 6) = 4.12\%$ for a correct model
- With $p = 0.03$: $P(X \leq 4) = 12.8\%$: an inaccurate model escapes one time in eight **BCBS (1996)**

A6: FZ0 Loss of Two Forecasts [Solved]

Task

- Five daily losses (%): 0.8, -1.1, 3.4, 2.3, -0.5;
 $\alpha = 0.025$
- Forecast A: $v = 2.0$, $e = 2.6$; forecast B: $v = 2.6$,
 $e = 3.4$ (constant, in %)
- (a) Breach days and the constant part $v/e + \ln e - 1$ of
each forecast
- (b) FZ0 on each day and the average; which forecast is
better?

Solution

- A: breaches on days 3 and 4; constant
 $0.769 + 0.956 - 1 = 0.725$
- B: breach on day 3 only; constant
 $0.765 + 1.224 - 1 = 0.988$
- A, day 3: $\frac{3.4-2.0}{0.025 \cdot 2.6} + 0.725 = 22.263$; day 4: 5.340; mean
5.956
- B, day 3: $\frac{3.4-2.6}{0.025 \cdot 3.4} + 0.988 = 10.400$; mean 2.871
- B is better: its higher VaR and ES cost little on quiet days
and save a lot on breach days



A7: Acerbi–Szekely Z_1 and Z_2 , Step by Step [Solved]

Task

- One year, $T = 250$, $\alpha = 0.025$; the VaR 2.5% was breached on 8 days
- Losses on those days (%):
3.9, 2.8, 4.6, 3.1, 5.2, 2.9, 3.6, 7.4
- Forecast ES on those days (%):
3.4, 3.3, 3.6, 3.2, 3.5, 3.3, 3.4, 3.8
- (a) Ratios L_t/ES_t ; (b) Z_1 and Z_2 ; (c) decision with the fixed thresholds

Solution

- Ratios:
1.147, 0.848, 1.278, 0.969, 1.486, 0.879, 1.059, 1.947; sum 9.613
- $Z_1 = 1 - 9.613/8 = 1 - 1.202 = -0.202$
- $Z_2 = 1 - 9.613/(250 \cdot 0.025) = 1 - 9.613/6.25 = -0.538$
- $Z_2 > -0.70$: green; 8 breaches instead of 6.25 and tail losses 20% above ES are not yet conclusive
- One more breach with $L = 7\%$ and $ES = 3.5\%$ would give $Z_2 = -0.86$: yellow; the statistic is driven by the largest losses

A8: Conformal Quantile and One ACI Step [Solved]

Task

- Calibration scores $s = L/\hat{\sigma}$ (sorted, $n = 19$): 0.19, 0.35, 0.52, 0.61, 0.73, 0.88, 0.94, 1.05, 1.12, 1.28, 1.47, 1.66, 1.83, 1.95, 2.02, 2.28, 2.41, 2.75, 3.10
- (a) Conformal VaR 10% for tomorrow if $\hat{\sigma} = 1.4\%$
- (b) ACI with $\gamma = 0.02$, $\alpha_1 = 0.10$: next level after a breach and after no breach
- (c) New VaR after a breach

Solution

- $k = \lceil 20 \cdot 0.9 \rceil = 18$: $s_{(18)} = 2.75$, VaR $= 1.4 \cdot 2.75 = 3.85\%$
- Breach: $\alpha_2 = 0.10 + 0.02(0.10 - 1) = 0.082$; no breach: $\alpha_2 = 0.10 + 0.02 \cdot 0.10 = 0.102$
- After a breach: $k = \lceil 20 \cdot (1 - 0.082) \rceil = \lceil 18.36 \rceil = 19$, VaR $= 1.4 \cdot 3.10 = 4.34\%$
- After no breach: $\lceil 17.96 \rceil = 18$: the VaR does not move; ACI moves in jumps of one order statistic

A9: Is a Larger ES Always Penalised? [Proposed]

Solution discussed in class

Task

- Same five losses as A6; forecast C: $v = 2.0$, $e = 4.0$
- (a) FZ0 on each day and the average for C
- (b) Rank A, B, C; explain why C beats A although its VaR is the same
- (c) Could a forecaster lower the average FZ0 by reporting $e < v$? Why is this excluded?
- Model: A6

B1: Backtesting Six VaR Models on the S&P 500 [Solved]

Task

- Rolling one-day VaR 1% forecasts of the six models, 2007–2026 ($T = 4959$)
- (a) Breaches, Kupiec p -value (asymptotic and exact Binomial)
- (b) Christoffersen independence and conditional coverage
- (c) Weibull duration test Christoffersen & Pelletier (2004)

Interpretation

- Which models pass all tests? Which test is decisive?
- When do asymptotic and exact p -values differ materially?
- Why can a model with the right number of breaches still fail?

B1: Results [Solved]

Model	x	Rate (%)	p_{uc}	p_{exact}	p_{ind}	p_{cc}	b / p_{dur}
HS	83	1.67	<0.001	<0.001	<0.001	<0.001	0.52 / <0.001
Normal	134	2.70	<0.001	<0.001	<0.001	<0.001	0.56 / <0.001
Student-t	99	2.00	<0.001	<0.001	<0.001	<0.001	0.53 / <0.001
GARCH-t	88	1.77	<0.001	<0.001	0.095	<0.001	0.94 / 0.479
FHS	61	1.23	0.116	0.127	0.046	0.040	0.93 / 0.514
GARCH-EVT	56	1.13	0.370	0.395	0.028	0.060	0.88 / 0.240

- Only GARCH-EVT passes Kupiec, conditional coverage and durations at 5%; FHS fails p_{cc} narrowly
- Durations are decisive for HS, Normal and Student-t ($b \approx 0.5$): clustered breaches
- With 4959 days asymptotic and exact p -values agree; in 2008 alone ($T = 253$, $x = 8$): 0.0058 versus 0.0086

B2: Traffic Light for GARCH-t on the BET [Proposed]

Solution discussed in class

Task

- ▣ VaR 1% of GARCH-t on the BET index, 2007–2026
- ▣ (a) Exceptions in the last 250 days; zone, cumulative probability and FRTB multiplier
- ▣ (b) Rolling 250-day counts: share of time in each zone; worst window
- ▣ (c) Desk check: exceptions of the VaR 2.5% in the last 250 days versus the limit of 30
- ▣ Model: A5 (zones), B1 (breach series)

Interpretation

- ▣ Would a supervisor have penalised this model at any time since 2007?
- ▣ Is the yellow zone evidence against the model or bad luck? Use the Binomial probabilities
- ▣ What does the traffic light ignore that Z_2 would capture?

B3: ES Backtests with Bootstrap and Simulation [Solved]

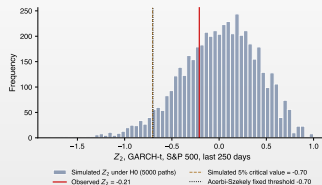
Task

- S&P 500, (VaR 2.5%, ES 2.5%) forecasts of GARCH-t, FHS and HS
- (a) McNeil–Frey: mean exceedance residual, t -statistic, bootstrap p -value (5000 resamples) versus Student- t p -value
- (b) Z_2 on the full sample and on the last 250 days, p -values from 5000 paths simulated from each model's forecast distributions

Interpretation

- Why is the bootstrap preferred for the exceedance residuals?
- Does the simulated 5% critical value for $T = 250$ agree with the fixed -0.70 ?
- Why is the full-sample critical value so much closer to zero?

B3: Results [Solved]



Model	n	$\bar{e} / t$	p_{boot} / p_t	$Z_2 (T = 4959)$	$p / \text{crit.}$	$Z_2 (250 \text{ d}), p$
GARCH-t	192	0.106 / 1.80	0.023 / 0.037	-0.618	<0.001 / -0.153	-0.210, 0.289
FHS	130	0.093 / 1.21	0.096 / 0.114	-0.091	0.154 / -0.150	0.142, 0.602
HS	156	0.363 / 2.88	<0.001 / 0.002	-0.405	<0.001 / -0.154	0.723, 0.975

- Bootstrap p -values are smaller than Student- t ones: residuals are skewed, the t approximation is poor with fat tails
- Full sample: GARCH-t and HS rejected; FHS passes ($p = 0.154$)
- Last 250 days: simulated 5% critical value -0.705 for GARCH-t, close to -0.70; nobody is rejected in a calm year

B4: Z_2 by Simulation on BET and Bitcoin [Proposed]

Solution discussed in class

Task

- (VaR 2.5%, ES 2.5%) forecasts of GARCH-t and GARCH-EVT on the BET and on Bitcoin
- (a) Z_2 on the full evaluation sample; p -value and 5% critical value from 5000 simulated paths
- (b) The same on the last 250 days; compare with the fixed threshold -0.70
- Model: B3

Interpretation

- Which model would you trust for ES on the BET, and on Bitcoin?
- Why does the verdict change between the full sample and the last year?
- How many breach days does the last-year test rely on?

B5: Diebold–Mariano with HAC Variance [Solved]

Task

- ▣ FZ0 losses of the (VaR 2.5%, ES 2.5%) forecasts on the S&P 500
- ▣ (a) Average FZ0 by model; best model
- ▣ (b) DM tests of GARCH-t, HS and GARCH-EVT against FHS with Newey–West variance (9 lags)
- ▣ (c) The naive t -test that ignores autocorrelation; ratio of long-run to ordinary variance

Interpretation

- ▣ Which differences are significant, and which are not?
- ▣ When does ignoring autocorrelation overstate significance?
- ▣ Is a significant FZ0 difference also economically large?

B5: Results [Solved]

- Average FZ0: FHS -3.551 , GARCH-EVT -3.548 , GARCH-t -3.512 , HS -3.134 , Student-t -3.036 , Normal -2.958
- GARCH-t minus FHS: $\bar{d} = 0.0388$, $DM = 2.65$ ($p = 0.008$); naive $t = 2.72$
- HS minus FHS: $DM = 4.70$ ($p < 0.001$) but naive $t = 7.18$; long-run variance 2.34 times the ordinary one
- GARCH-EVT minus FHS: $DM = 1.24$ ($p = 0.216$): no significant difference
- HS losses cluster in crises, so its loss differential is autocorrelated: the naive test overstates significance

 MFM_ch8_sem_compare

B6: Comparing Models on Bitcoin and EUR/RON [Proposed]

Solution discussed in class

Task

- FZ0 losses of the six models on Bitcoin and on EUR/RON
- (a) Best model; DM tests of every other model against it (HAC variance)
- (b) MCS p -values (T_{max} statistic, block bootstrap, 1000 replications) Hansen, Lunde & Nason (2011)
- Model: B5

Interpretation

- Why is the MCS so large for Bitcoin and so small for EUR/RON?
- Relate the answer to the effective tail sample αT Pele & Mazurencu-Marinescu-Pele (2026)

B7: Conformal VaR in Calm and Crisis Periods [Solved]

Task

- S&P 500; score $s_t = L_t / \hat{\sigma}_t$, $\hat{\sigma}_t$ from the previous 250 returns; calibration on the last 250 scores
- (a) Split conformal VaR 5% and VaR 1%, 2007–2026: breach rates overall, in calm and in crisis days (previous 20-day volatility in its top 10%)
- (b) Repeat on simulated i.i.d. Student- t_4 data of the same length (exchangeable)
- (c) Christoffersen independence test on both breach series

Interpretation

- Is the marginal guarantee met on real data?
- What changes between real and exchangeable data: coverage or independence?

B7: Results [Solved]

	All	Calm	Crisis	p_{uc}	p_{ind}	i.i.d.: all / calm / crisis
VaR 5%	5.34	5.15	7.06	0.272	<0.001	4.74 / 4.73 / 4.84
VaR 1%	1.29	1.21	2.02	0.049	0.001	0.73 / 0.76 / 0.40

- ▣ Breach rates in %; crisis = the 496 days whose previous 20-day volatility is in the top 10%
- ▣ Marginal coverage almost holds (VaR 5%: 5.34%), but crisis days breach at 7.06% and breaches cluster ($p_{ind} < 0.001$)
- ▣ i.i.d. data: calm 4.73%, crisis 4.84%, independence holds ($p_{ind} = 0.965$): the guarantee works as promised
- ▣ Lesson: the guarantee is marginal; volatility clustering breaks independence of breaches

B8: Conformal VaR on BET and Bitcoin [Proposed]

Solution discussed in class

Task

- VaR 5% on BET and Bitcoin: split conformal, ACI ($\gamma = 0.005$) and GARCH-t
- (a) Breach rates overall, calm and crisis for each method
- (b) Which method is closest to 5% in crises?
- Model: B7

Interpretation

- Does ACI fix conditional coverage, or only the long-run average?
- Would you use a conformal VaR for regulatory capital? Why?

C1: Which Risk Model Would You Certify for a Romanian Bank's Trading Book? [Proposed]

Solution discussed in class

The question

- A trading book in lei: half in the BET index, half in euro (valued at the BNR reference rate)
- As model validator, you must certify one VaR/ES model for FRTB reporting
- Combine regulatory checks (traffic light, desk limits) with statistical evidence (tests, FZ0, MCS)

Deliverable

- A Quantlet folder in the team repository and one slide with the recommendation and its main limitation

Hints

- Join the two price series on common days first, then compute returns
- Use at least three of: Kupiec, Christoffersen, durations, Z_2 , FZ0 with DM tests and MCS
- Look at the worst 250-day window, not only the last one
- Would your choice survive a year like 2008 or 2020?

C2: Critique an AI Answer (1/3) [Proposed]

Solution discussed in class

The student asks an AI assistant

- "I have daily S&P 500 closes in a pandas Series P. Write Python for a rolling 250-day historical-simulation VaR 1%, count the exceptions and run the Kupiec and Christoffersen tests."
- "My last 250 days have 7 exceptions. What does that mean for the model?"

The AI answer: text

- Losses $L_t = -r_t$; VaR_t is minus the 1% quantile of the last 250 returns; exception: $L_t > \text{VaR}_t$
- `lr_ind(I)` returns the Christoffersen independence statistic LR_{ind} Christoffersen (1998)
- Your last 250 days: $x = 7$, $\text{LR}_{uc} = 5.50$, $p = 0.019$: correct coverage is rejected at 5%
- Basel traffic light: 7 exceptions is the red zone, so the model must be withdrawn BCBS (1996)

C2: Critique an AI Answer (2/3) [Proposed]

Solution discussed in class

The AI answer: the code that came with the text

```
import numpy as np, scipy.stats as stats
r = np.log(P).diff().dropna().values
W, p = 250, 0.01
VaR = np.array([-np.quantile(r[t-W:t+1], p)
                 for t in range(W, len(r))])
I = (-r[W:] > VaR).astype(int)
T, x = len(I), I.sum()
ph = x / T
LR_uc = -2*((T-x)*np.log(1-p) + x*np.log(p)
           - (T-x)*np.log(1-ph) - x*np.log(ph))
LR_cc = LR_uc + lr_ind(I)
p_uc = stats.chi2.sf(LR_uc, df=1)
p_cc = stats.chi2.sf(LR_cc, df=1)
```

C2: Critique an AI Answer (3/3) [Proposed]

Solution discussed in class

Task

- ▣ The answer contains three errors; everything else is correct: find them
- ▣ For each error, say how you would detect it: a test, a published number, the lecture
- ▣ Give the corrected version: formula, code line and number
- ▣ Log each error as in `AI_ERRORS.md`
 - ▶ What you asked, what the AI answered, what was wrong, how you found out, the correct version

Hints

- ▣ Which returns are known at the close of day $t - 1$, when VaR_t is reported?
- ▣ How many restrictions does the conditional coverage test impose?
- ▣ Compare the zone with A5 and with the Basel table [BCBS \(1996\)](#)



Key Takeaways

- ▣ Kupiec counts, Christoffersen and durations look at timing: always use both
- ▣ The traffic light is a Binomial test; it can let an inaccurate model through one time in eight
- ▣ ES tests need the forecast distribution: bootstrap and simulation are the honest way to get p -values
- ▣ Rank with FZ0 + DM (HAC) + MCS; significance needs many tail days
- ▣ Conformal coverage is marginal; in crises it fails like an unconditional model

Take-Home Assignment (Team Project)

- Choose a portfolio not covered in class (a BVB stock, a currency or a crypto-asset); forecast VaR 1% and ES 2.5% with at least three models
- Backtest them with Kupiec, Christoffersen, durations and Z_2 ; place each model in the traffic light for its worst 250-day window
- Rank them with FZ0, Diebold–Mariano tests (HAC) and the MCS
- Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- Log every AI error you catch (at least three per project) in `AI_ERRORS.md` and declare your AI use in `AI_USE.md`

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